

The Data Layer

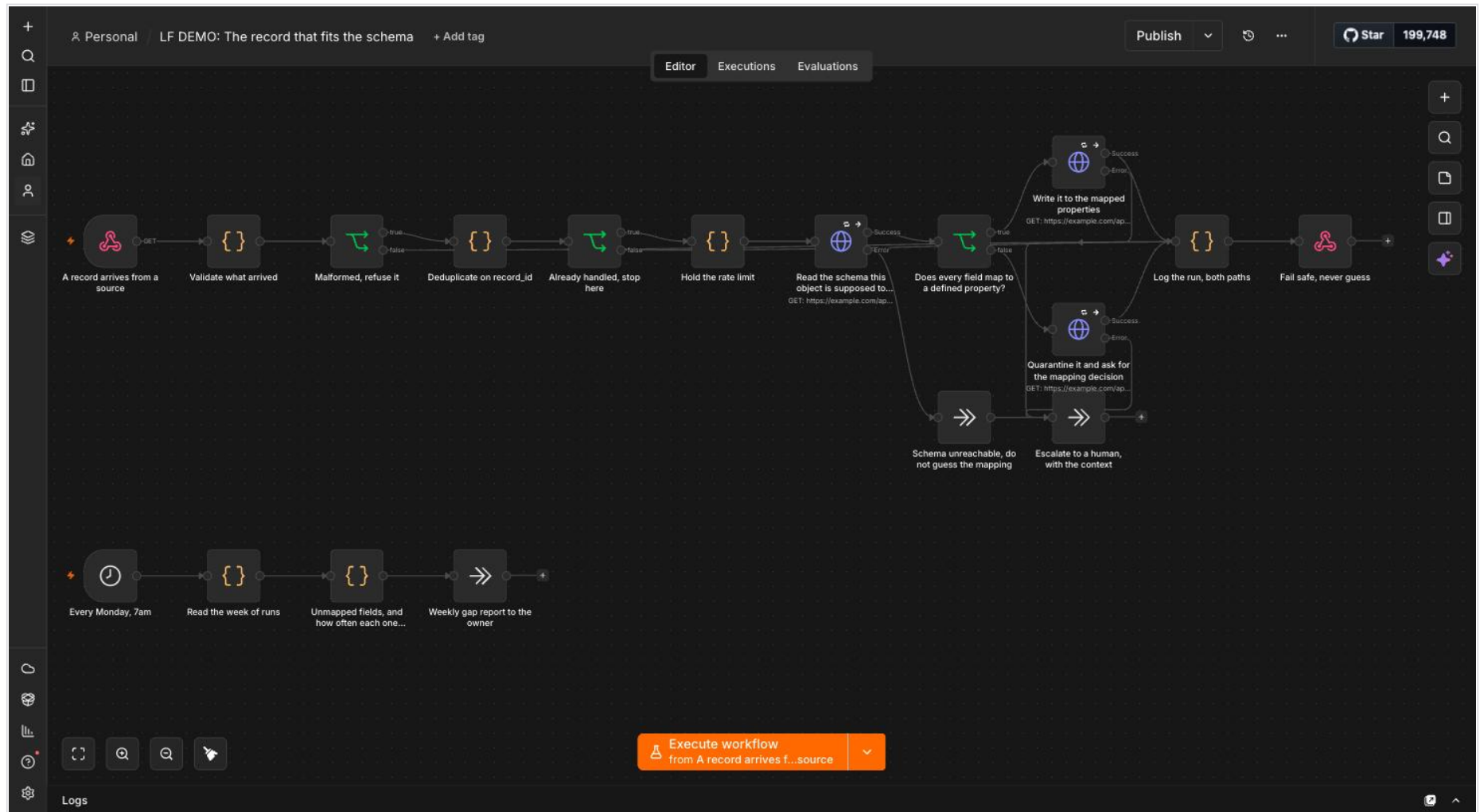
data-model-architecture · built in hubspot

Most CRM problems are data-model problems wearing a workflow costume.

What it connects to: The schema has to hold for every system that writes into it. A property that only makes sense in one tool is the one that breaks the reporting later.

1 of 4 SYSTEM

Here's the system that enforces it — a record only gets written once every field maps to a property that exists.



2 of 4 THINKING

This is the gate that keeps the model clean: every field has to map to a property that exists, or the record is quarantined rather than guessed at. Guessing is how you end up with three fields meaning the same thing.

The screenshot shows a dark-themed user interface for a data processing tool. The main window is titled "Does every field map to a defined property?". The interface is divided into two main sections: "INPUT" on the left and "Parameters" / "Settings" on the right.

INPUT Section:

- At the top, it says "INPUT".
- Below that, there is a large area with the text "No input data" and a button "Execute previous nodes" with the subtitle "to view input data".
- On the far left, there is a vertical sidebar with various icons, including a globe icon.

Parameters / Settings Section:

- At the top right, there is an orange button labeled "Execute step".
- Below the button, there are two tabs: "Parameters" (selected) and "Settings".
- Under the "Parameters" tab, there is a section titled "Conditions".
- Inside the "Conditions" section, there is a text area containing the following code:

```
{{ $json.fullyMapped === true && $json.objectExists === true }}
```
- To the right of the code area, there is a checkbox labeled "is true" which is checked.
- Below the code area, there is a button labeled "+ Add condition".
- Below the "Conditions" section, there is a toggle switch labeled "Convert types where required" which is currently turned off.
- Below the toggle switch, there is a section titled "Options" with a button labeled "+ Add option".

3 of 4 THEIR-WORLD

Here's the object model on a real portal — objects, properties, and how they associate.

The screenshot displays the 'Data Model' section of a portal, specifically the 'Manage' tab. The interface is organized into several key areas:

- Top Bar:** Includes a 'Find or Ask' search bar, navigation icons for Upgrade, Assistant, and Launch Forte, and buttons for 'Self-paced tours' and 'Import'.
- Left Sidebar:** Labeled 'Objects', it contains a search bar and a list of object categories: All objects, Calls, Carts, Communications, Companies, Contacts, Contracts, Credit memos, Deals, Emails, Invoices, and Line items.
- Main Content Area:** Titled 'Viewing all objects', it features a search bar and a dropdown for '(2) Object status'. Below this is a grid of object cards, each representing a different entity type with its associated properties and linked objects.

The object cards are categorized into three groups:

- CRM objects:**
 - Appointments:** 47 properties, 14 associated objects.
 - Carts:** 47 properties, 14 associated objects.
 - Companies:** 118 properties, 21 associated objects.
 - Contacts:** 214 properties, 20 associated objects.
 - Courses:** 70 properties, 20 associated objects.
 - Listings:** 70 properties, 20 associated objects.
 - Marketing events:** 25 properties, 12 associated objects.
 - Orders:** 70 properties, 20 associated objects.
 - Projects:** 48 properties, 12 associated objects.
 - Services:** 40 properties, 18 associated objects.
 - Tickets:** 40 properties, 18 associated objects.
 - Users:** 48 properties, 12 associated objects.
- Sales objects:**
 - Contracts:** 80 properties, 18 associated objects.
 - Credit memos:** 59 properties, 6 associated objects.
 - Deals:** 74 properties, 22 associated objects.
 - Invoices:** 91 properties, 12 associated objects.
 - Line items:** 55 properties, 10 associated objects.
 - Payments:** 92 properties, 11 associated objects.
 - Quotes:** 89 properties, 12 associated objects.
 - Subscriptions:** 41 properties, 11 associated objects.
- Activities:**
 - Calls:** 43 properties, 8 associated objects.
 - Communications:** 19 properties, 8 associated objects.
 - Meetings:** 37 properties, 8 associated objects.
 - Notes:** 17 properties, 8 associated objects.

Each object card includes an 'Activate object' button and a 'View all' link. A 'Create object' button is also present in the top right of the main content area.

And the property architecture underneath it, with fill rates so you can see what's actually used.

Find or Ask

Upgrade

Assistant

Launch Forte

Dashboard

Search Settings

Notifications

Account Management

Account Defaults

Account & Billing

Audit Log

Users & Teams

Product Updates

Integrations

Marketplace Downloads

Tracking & Analytics

Privacy & Consent

Security

AI

Payments Account

Data Management

Properties

Properties

Data Agent

Data Quality

Export properties

Properties are used to collect and store information about your records in HubSpot. For example, a contact might have properties like First Name or Lead Status.

Select an object:

Contact properties

Go to Contacts settings

Properties (211)

Conditional logic

Groups

Archived (0)

All groups

All field types

All users

All properties

Search properties

Data quality monitoring is off

Create property

	Name ↑	Group T↓	Created by T↓	Used In T↓	Fill Rate
☐	Annual Revenue Single-line text	Contact information	HubSpot	0	0%
☐	Average Pageviews Number	Web analytics history	HubSpot	0	100%
☐	Buying Role Multiple checkboxes	Deal information	HubSpot	0	0%
☐	Campaign of last booking in meetings tool Single-line text	Contact activity	HubSpot	0	0%
☐	Chat Assistant IQL Date Date and time picker	Contact information	HubSpot	0	0%